



8170 - The power of the dark side: Weighing a super-massive neutron star with JWST

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec Fixed Slit Spectroscopy	(1) PSR J2215+5135

ABSTRACT

JWST spectroscopy provides a unique opportunity to accurately measure the mass of the heaviest known pulsars, within compact binary millisecond pulsars (also known as redbacks and black widows). The maximum mass that a neutron star can support, M_{max} , has important implications across multiple research fields, including astrophysics, nuclear physics and gravitational wave astronomy. In 2018, ground-based observations revealed one of the most massive neutron stars in the redback pulsar PSR J2215+5135, with 2.3 Solar masses (M_{sun}). This gave tantalizing evidence that M_{max} is higher than 2.3 M_{sun} , yet the relatively large (7%) uncertainty prevents firm conclusions. This GO program capitalizes on the superior NIR spectroscopic capabilities of JWST-NIRSpec in order to improve the accuracy of this mass measurement, down to 2%. JWST observations at 1-2 microns can measure the velocity and temperature of both the dark and bright sides of the irradiated companion star. Establishing with high

confidence the existence of super-massive neutron stars, well above 2 Msun, would have profound consequences for our understanding of matter at supra-nuclear densities ($>3E14 \text{ g/cm}^3$).

OBSERVING DESCRIPTION

We request the first JWST observations of the compact binary "redback" millisecond pulsar PSR J2215+5135, in order to measure its mass with 2% accuracy. Our goal is to obtain high-quality ($S/N > 30$) and high-resolution ($R = 2700$) NIR spectra of the companion star throughout an entire orbit, using NIRSpec on the JWST. This will allow us to i) measure radial velocities with unprecedented precision; and ii) measure the temperature of the companion throughout the orbit, which shows large variations due to irradiation by the pulsar wind.

Proposal 8170 - Targets - The power of the dark side: Weighing a super-massive neutron star with JWST

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	PSR_J2215+5135	RA: 22 15 32.6869 (333.8861954d)	Proper Motion RA: 0.014 mas/yr	
			Dec: +51 35 36.40 (51.59344d)	Proper Motion Dec: 2.240 mas/yr	
			Equinox: J2000	Epoch of Position: 2015.5	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Coordinates (J2000) and proper motions (J2015.5) have been updated to Gaia EDR3 (minor ~0.1% changes with respect to APT/SIMBAD's query results).</i> <i>Category=Star</i> <i>Description=[Pulsars]</i>				

Proposal 8170 - Observation 1 - The power of the dark side: Weighing a super-massive neutron star with JWST

Observation	Proposal 8170, Observation 1										Thu May 21 17:00:20 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRSpec Fixed Slit Spectroscopy										
	(Visit 1:1) Warning (Form): Data Excess over lower threshold										
Fixed Targets	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
Acquisition	(1)	PSR_J2215+5135	RA: 22 15 32.6869 (333.8861954d)			Proper Motion RA: 0.014 mas/yr					
			Dec: +51 35 36.40 (51.59344d)			Proper Motion Dec: 2.240 mas/yr					
Template			Equinox: J2000			Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
Dithers	Coordinates (J2000) and proper motions (J2015.5) have been updated to Gaia EDR3 (minor ~0.1% changes with respect to APT/SIMBAD's query results).										
	Category=Star										
Dithers	Description=[Pulsars]										
	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
Dithers	1	1 PSR_J2215+5135	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	146180
Dithers	HFF Readout Mode										
	Slit										
Dithers	Subarray										
	false										
Dithers	S200A1										
	FULL										
Dithers	#	Primary Dither Positions					Sub-Pixel Pattern				
	1	3					NONE				

Proposal 8170 - Observation 1 - The power of the dark side: Weighing a super-massive neutron star with JWST

Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	1	NONE	3	6	962.867	
	2	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	2	NONE	3	6	962.867	
	3	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	3	NONE	3	6	962.867	
	4	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	4	NONE	3	6	962.867	
	5	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	5	NONE	3	6	962.867	
	6	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	6	NONE	3	6	962.867	
	7	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	7	NONE	3	6	962.867	
	8	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	8	NONE	3	6	962.867	
	9	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	9	NONE	3	6	962.867	
	10	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	10	NONE	3	6	962.867	
	11	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	11	NONE	3	6	962.867	
	12	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	12	NONE	3	6	962.867	
	13	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	13	NONE	3	6	962.867	
	14	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	14	NONE	3	6	962.867	
	15	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	15	NONE	3	6	962.867	
	16	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	16	NONE	3	6	962.867	
	17	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	17	NONE	3	6	962.867	
	18	G140H/F070LP	S200A1	NRSIRS2RAPID	10	2	18	NONE	3	6	962.867	